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The Palaeomagnetism of the Umkondo Lavas, Eastern Southern Rhodesia

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Summary

The mean direction of magnetization of samples collected from nine sites in the Umkondo lavas shows no significant difference from that of the Umkondo dolerites after demagnetization in alternating magnetic fields. This supports the contention of geologists that the dolerites are the intrusive phase of the lavas. The lavas exhibit considerably greater dispersion than the dolerites and it seems possible that the sampling of the latter may have been insufficient to average out secular variation. The combined data for the lavas and the dolerites gives an estimate of the angular dispersion of the geomagnetic field in Umkondo times which is in good agreement with that calculated for the present field.

1. Introduction

The Umkondo System lies southeast of Southern Rhodesia on the border of Moçambique. Descriptions of various parts of the Umkondo System have been given by Swift *et al.* (1953), Swift (1962) and Watson (1964). It rests upon the basement granite of the Rhodesian Shield and is presumed to be Pre-Cambrian in age. In the extreme southeast it is overlain unconformably by beds of the Karroo System. Since it is completely isolated from other Pre-Cambrian Systems in Rhodesia its age has thus been a matter of some speculation.

Fig. 1 shows a geological map of the Umkondo System, which consists of the Calcareous Series composed of a basal arkose and limestones about 800 ft thick. This is overlain by the Lower Argillaceous Series of grey shales and siltstones about 800 ft thick, followed by the Quartzite Series of massive dark grey crystalline quartzitic sandstones some 3 000 ft thick. The highest sequences consist of the Upper Argillaceous Series of pale and dark purple shales and siltstones about 1 000 ft thick and a volcanic series of basic, often amygdaloidal, lavas. As shown in Fig. 1, these are preserved in isolated patches, which have been named, as indicated, by the Rhodesian Selection Trust Company's mineral exploration geologists in the Sabi Valley region and author in the areas bordering Moçambique in the extreme southeast. The Umkondo rocks have been intruded by large quantities of dolerite. The main sill is some 600 ft thick intruding the Calcareous Series. Smaller sills intrude higher horizons and in places this dolerite is petrologically identical

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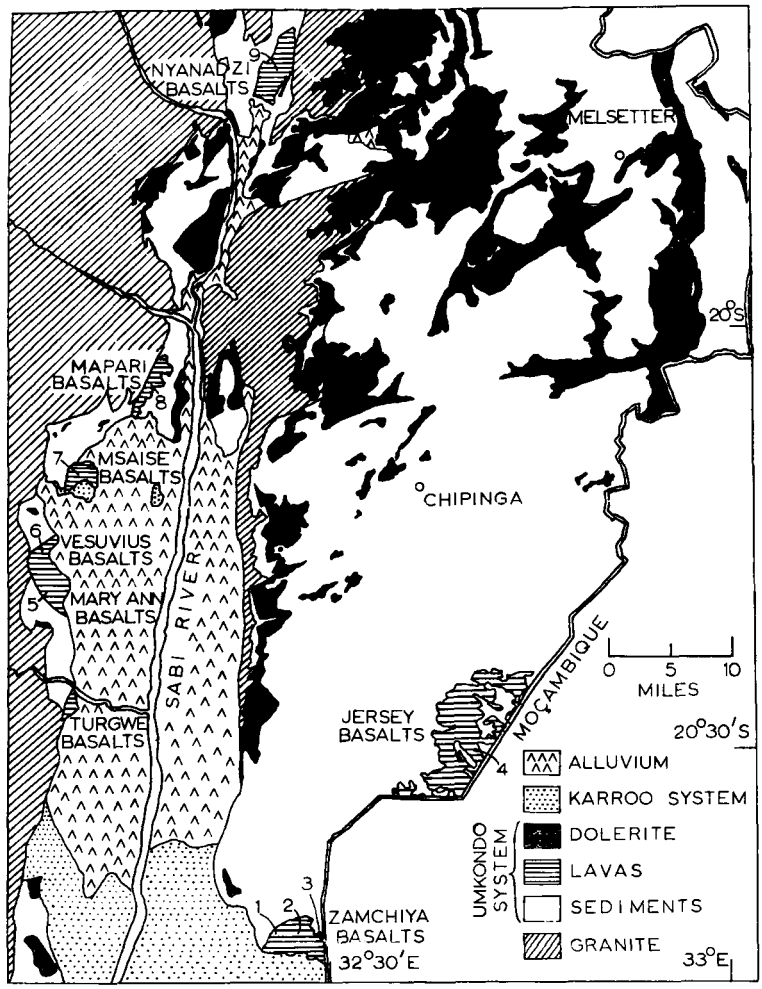


FIG. 1. Geological map of southeast Southern Rhodesia showing the Umkondo System and the location of the sampling sites.

with that of the main sill. Elsewhere it becomes granophyric by digestion of feldspathic sandstone.

The Umkondo System shows evidence throughout of shallow water deposition. The limestones often show oölitic texture and the succeeding groups are frequently ripple-marked throughout the sequence.

2. The Umkondo lavas and dolerites

The lavas attain a thickness of 600 ft or more and are very poorly exposed. Both amygdaloidal and non-amygdaloidal varieties are seen but it is not possible to distinguish separate flows. The Nyanyadzi basalts have been found interbedded with the upper quartzites and elsewhere thin quartzitic bands are seen in the lavas. The lavas are hence conformable in the sequence and belong to the Umkondo System. Both Swift (1962) and Watson (1964) suggest that the dolerite sills represent the intrusive phase of the lavas.

The so-called basalts, which are basic andesites, are frequently chocolate coloured, usually with green chloritic amygdales. In thin sections the rocks are seen to consist of small crystals of feldspar and pyroxene in a blown glass which is

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devitrified to varying degrees. The pyroxene is an almost colourless augite, usually chloritized; bastite pseudomorphs also occur. The feldspar is probably andesine, but is highly sericitized.

In a previous investigation, McElhinny & Opdyke (1964) made a palaeomagnetic study of the Pre-Cambrian dolerites of eastern Southern Rhodesia. They found that the dolerites represented two distinct magmatic episodes, the Umkondo dolerites showing an entirely different direction of magnetization from the Mashonaland dolerites, which intrude the Rhodesian Shield to the west of the Umkondo System. At one locality the Mashonaland dolerites were found apparently intruding the Umkondo dolerites, which made it possible to determine their relative ages. The Mashonaland dolerites have been dated at 1600 million years as reported by McElhinny & Opdyke (1964), which would therefore also indicate that the Umkondo System is at least this age.

Attempts are at present being made to obtain isotopic dates from samples of Umkondo dolerite. Since the dolerites are regarded as the intrusive phase of the lavas such dates will therefore be held to refer to the age of the Umkondo System itself. For this reason it was important to obtain some confirmation of this hypothesis. Such was the main purpose of the present investigation.

It is hoped that palaeomagnetic pole positions might be used in the future as a means of correlating isolated Pre-Cambrian Systems in Southern Africa. Confirmation that the pole position for the Umkondo dolerites refers also to the Umkondo System itself would be invaluable as a first step in this direction.

3. Sampling and measurement

Since the Umkondo lavas are very poorly exposed, good outcrops were often very hard to find. All except one of the isolated exposures of the lavas were sampled (Fig. 1). The Turgwe basalts were inaccessible and were not sampled. In the Sabi Valley the few outcrops were so small that it was often only possible to drill a single 2 ft core using a Graham–Keiller type sampling drill (Graham & Keiller 1960). The nine sampling sites are indicated in Fig. 1 and at four of these only a single 2 ft core could be obtained. The Mary Ann basalts were sampled underground in the Umkondo Copper Mine and three blocks, oriented with a compass, were collected. In all other cases orientation was with respect to the Sun's azimuth using the method described by Gough & Opdyke (1963).

At Zamchiya, on the Moçambique border, it was possible to sample over a thickness of 350 ft. Site 2 is about 100 ft above site 1, and site 3 about 250 ft above site 2. Site 4, in the Jersey basalts, was previously sampled by McElhinny & Opdyke (1964) as Umkondo dolerite and thought to be a very fine grained variety. At the time of sampling, geological mapping in the area was still in progress. Mapping further south the following field season by Watson (1964) established that these were in fact the northerly extension of the Zamchiya basalts. The connection between the two was traced across the Moçambique border and back into Southern Rhodesia. The Zamchiya basalts often contain amygdales, whereas the Jersey basalts are the non-amygdaloidal variety. Site 4 is therefore that reported as site J by McElhinny & Opdyke (1964) and the measurements made on seven 6 in. cores at this locality are reported here again (see Table 1).

The natural remanent magnetization of the samples at each of these sites was measured on the spinner magnetometer described by Gough (1964). Two specimens were cut from each core and the mean of these was regarded as a single observation.

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4. Demagnetization of the Umkondo lavas

The samples from each of the sites were successively demagnetized in the apparatus described by McElhinny (1965) and using the techniques previously described by McElhinny & Gough (1963). The tumbler having a ratio of 15:16 used by these latter authors was used in this investigation, the 11:16 ratio tumbler only being installed later. Almost all the samples were extremely stable in high fields of up to 100 mWb m^{-2} peak value. In most cases a small viscous component in the direction of the present Earth's field was removed in 10 or 20 mWb m^{-2} and the specimens exhibited the typical behaviour of those with stable thermoremanent magnetization after demagnetization in higher fields. Table 1 gives the directions of magnetization at each site before and after demagnetization. The between site precision parameter k (Fisher 1953) is 3·1 for the NRM and this rises to 12·8 after demagnetization.

Table 1

Site directions of magnetization for the Umkondo lavas

Site no.	Area	Position Lat. S	Position Long. E	N	B_{AC}	M_{AC} D	M_{AC} I	Site pole Lat.	Site pole Long.
1	Zamchiya basalts	20° 42′	32° 28′	2	25	173	−11	63 N	17 E
2	Zamchiya basalts	20° 42′	32° 29′	1	30	178	−8	65 N	27½ E
3	Zamchiya basalts	20° 43′	32° 30′	3	40	159	−19	53 N	3½ W
4	Jersey basalts	20° 32′	32° 40′	7	80	180½	−10	64½ N	34 E
5	Mary Ann basalts	20° 19′	32° 09′	3	80	200	−5	60 N	75 E
6	Vesuvius basalts	20° 15′	32° 09′	1	50	177	+14	76½ N	45 E
7	Msaise basalts	20° 10′	32° 11′	1	20	169	+6	69½ N	65½ E
8	Mapari basalts	20° 04′	32° 17′	2	80	142	−36	35½ N	10½ W
9	Nyanyadzi basalts	19° 42′	32° 27′	1	40	164	−38	46 N	23½ W

N = No. of cores.
B_{AC} = Peak demagnetizing field in mWb m^{-2}.
M_{AC} gives the mean direction of magnetization after demagnetization in the field B_{AC} and D and I are the declination and inclination of this direction.

The directions have been corrected for the attitude of the lavas at each site. Generally the lavas dip at about 10° in an easterly or southeasterly direction, except at site 5 where they dip at 35° to the northeast. Fisher's k for the directions before applying these corrections is 10·2 and this rises to 12·8 on restoring the lavas to the horizontal at each site.

For nine sites this does not represent a significant change in k (McElhinny 1964), but as the corrections are very small, this is hardly to be expected.

5. Relationship between the lavas and the dolerites

Fig. 2 shows the final site directions of magnetization together with the circle of 95% confidence for the mean. On the same stereographic projection is shown the circle of 95% confidence for the mean direction of magnetization of the Umkondo dolerites (McElhinny & Opdyke 1964). The agreement between the two is very good and no significant difference can be claimed. This result supports the contention of geologists that the dolerites are the intrusive phase of the lavas.

It was pointed out by Gough, Opdyke & McElhinny (1964) that the angular dispersion Δ defined as $\Delta = 81/k^{\frac{1}{2}}$ for the Umkondo dolerite directions was apparently significantly lower than that of the present Earth's field as analysed

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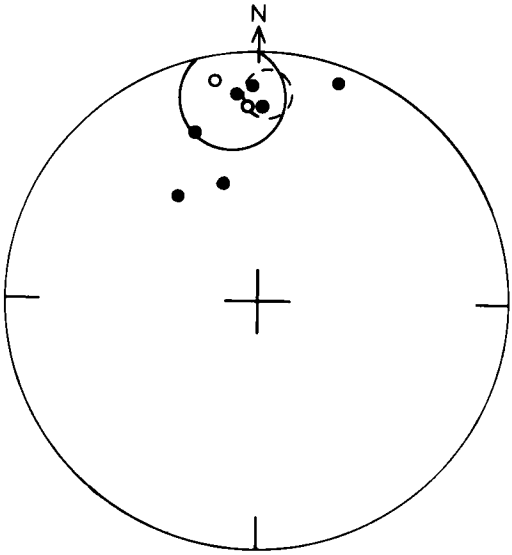


FIG. 2. Stereographic projection of the directions of magnetization at nine sites in the Umkondo lavas. ●, S-seeking direction on lower hemisphere; ○, S-seeking direction on upper hemisphere. ———, Circle of 95% confidence for the mean direction of the Umkondo lavas; – – – – – –, circle of 95% confidence for the mean direction of the Umkondo dolerites.

by Creer (1962). It is possible that this difference arises from inadequate sampling of the secular variation. Of the ten sites measured by McElhinny & Opdyke (1964) in the Umkondo dolerites, five in the Melsetter–Chipinga region (see Fig. 1) are regarded by Watson (1964) as being from the same sill, about 600 ft thick, which intrudes the Calcareous Series. Four of the remainder from the Inyanga region 100 miles to the north are also regarded as being from the same sill about 600 ft thick. According to Jaeger (1957), a 600 ft sill intruded at 1 100 °C into country rock at 0 °C will cool in such a way that the centre reaches 600 °C about 1 000 years later than the margins. It is therefore possible that sampling two such sills has resulted in the sites representing an insufficient length of time to average out secular variation.

There is no way of judging the time covered by the extrusion of the Umkondo lavas. It is not possible to distinguish individual flows, but the spread of the sites over the large area sampled and the difference in topographic position of the sites lends one to suppose that each site at least represents a different point in time. The eighteen sites sampled in both the lavas and the dolerites when combined together might therefore provide a somewhat better estimate of the angular dispersion in Umkondo times than either taken separately. When both the lavas and the dolerites are combined the parameters for the mean direction of magnetization of the eighteen sites are given in Table 2 together with the pole position for this mean. Alternatively the eighteen site poles can be combined to give the mean pole position; these parameters are also given in Table 2.

The estimate k of the between site precision κ for the eighteen directions of magnetization is 20, which corresponds to an angular dispersion of 18·1°. The use of an F–test, as described by Gough *et al.* (1964), gives upper and lower limits k_1 and k_2 such that there is 95% confidence that $k_2 \leqslant \kappa \leqslant k_1$. This corresponds to an upper limit of 24·2° and a lower limit of 13·6° in Δ. Creer (1962) estimates

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the angular dispersion for a latitude of 6° for the present Earth's field to be 18·5° which is in good agreement with that calculated from the combined data for the lavas and dolerites.

Table 2

(a) *Pole positions calculated from the mean directions of magnetization*

Unit	N	D	I	R	k	α_{95}	Lat.	Long.	δp	δm
Umkondo lavas	9	$172\frac{1}{2}°$	$-12\frac{1}{2}°$	8·377	12·8	15°	$62\frac{1}{2}°$ N	16° E	8°	15°
Combined lavas and dolerites	18	178°	$-11\frac{1}{2}°$	17·14	20	8°	64° N	28° E	4°	8°

(b) *Pole positions calculated by combining the site poles*

Unit	N	R	K	A_{95}	Lat.	Long.
Umkondo lavas	9	8·473	15·2	$13\frac{1}{2}°$	$63\frac{1}{2}°$ N	$16\frac{1}{2}°$ N
Combined lavas and dolerites	18	17·25	22·6	$7\frac{1}{2}°$	65° N	30° E

N is the number of sites, R is the length of the resultant of the N unit vectors and D and I are the declination and inclination of its direction. k is the precision of the site directions and K the precision of the site poles. α_{95} and A_{95} are the semiangle of the cone of 95% confidence for the site directions and site poles respectively, and δp and δm are the semiaxes of the oval of 95% confidence for the mean pole position.

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